

1. Consider a simple undirected graph G with 10 vertices and $\binom{10}{2}$ edges.

(a) Identify G and determine the degree of every vertex.

(b) Suppose an edge connecting vertices u and v is removed. What are the new degrees of u and v ?

(c) How many triangles are destroyed by removing the edge (u,v) ? Explain.

$$[2 + 1 + 2 = 5]$$

2. Prove that the number of triangles in an undirected graph is given by

$$N_{\Delta} = \frac{1}{6} \text{Tr}(A^3),$$

where A is the adjacency matrix of the graph.

Hints: Ensure that you do not over count.

[5]